Losing iron sites or losing the nickel host? Why Ni–Fe oxygen-evolution catalysts deactivate

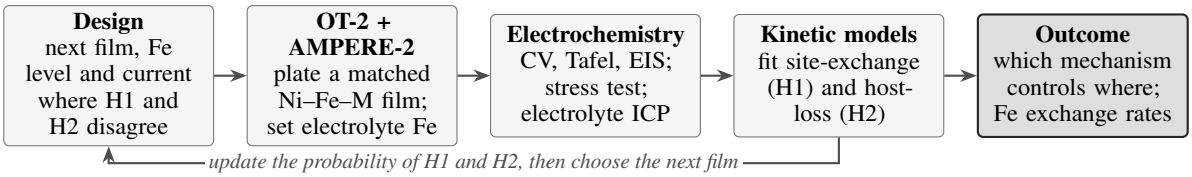
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Research question. Ni–Fe (oxy)hydroxides are among the most active earth-abundant catalysts for the oxygen evolution reaction (OER) in alkaline and anion-exchange-membrane water electrolysis, which avoids the iridium that acidic electrolysis needs [1,2]. Their activity fades under load. Is that because Fe active sites leave faster than they return, or because the Ni host itself transforms?

Knowledge gap. Fe in these catalysts is not fixed: it dissolves and redeposits during OER, and activity holds only while the two rates balance [3]. That has been shown for a few compositions. Whether site exchange, rather than restructuring of the host, controls durability across compositions, and how a third metal shifts the balance, has not been tested systematically, because each test consumes a film and needs matched electrolyte analysis.

Hypotheses. *H1, site exchange:* retained activity is set by the steady-state Fe site coverage, fixed by a composition-dependent Fe dissolution rate and an electrolyte-Fe-dependent redeposition rate. Predictions: adding Fe to the electrolyte restores activity, and Fe lost to the electrolyte predicts the activity decay of every composition with the same two rate laws. *H2, host restructuring:* activity loss follows irreversible change in the Ni host (fewer redox-active Ni sites, rising charge-transfer resistance, loss of film). Predictions: added Fe does not restore activity, and decay tracks the Ni redox charge (CV) and impedance (EIS), not electrolyte Fe.

Approach. An Opentrons OT-2 fitted with the open AMPERE-2 electrodeposition hardware [4] plates matched Ni–Fe–M films (Fe 0–40%; M = Co or Mn, 0–20%) on identical substrates by one fixed protocol: the films are a controlled variable, not the product. Each film gets CV, a Tafel scan and EIS, then a short stress test (constant current or potential cycling) in Fe-free KOH [2] with Fe added at set levels, while the robot draws electrolyte aliquots for ICP (Fe, Ni, M). Both mechanisms are written as kinetic models with shared parameters. Bayesian experimental design picks the next composition, electrolyte Fe level and current density where the models' predicted activity retention differs most, so AI serves mechanism identification.



Expected outcomes. (1) A map of where in composition and electrolyte space each mechanism controls durability; (2) Fe dissolution and redeposition rate constants as functions of host composition; (3) whether a third metal works by slowing Fe loss or by stabilizing the host. Together these explain how to keep earth-abundant active sites in place, which is what limits alkaline and anion-exchange-membrane electrolyzers.

Fit and resources. The project addresses catalyst *dynamics during catalytic cycles and deactivation*, *mechanisms mediated by earth-abundant metals*, and *AI/ML for mechanism identification*. It does not optimize catalyst synthesis or electrolyzer design. The PI works on self-driving labs and Bayesian experimental design; the lab has an OT-2, is adding a potentiostat, and has campus ICP access.

References. [1] M. W. Louie, A. T. Bell, *J. Am. Chem. Soc.* 135, 12329 (2013), doi:10.1021/ja405351s. [2] L. Trotochaud *et al.*, *J. Am. Chem. Soc.* 136, 6744 (2014), doi:10.1021/ja502379c. [3] D. Y. Chung *et al.*, *Nat. Energy* 5, 222 (2020), doi:10.1038/s41560-020-0576-y. [4] Fisker-Bødker *et al.*, *Digital Discovery* 4, 2491 (2025), doi:10.1039/D5DD00180C.